

StudentUtilityProgram

Java Programming Laboratory Experiment

Experiment No.	01
Date	11/09/2026
Title	Student Utility Program using Java

Objective / Aim

To design and implement a Java-based Student Utility Program using the Scanner class, switch statements, nested switch statements, conditional statements, loops, arithmetic operations, and user input. The program contains four parts covering student information and marks, decision-making problems, iterative operations, and mathematical programs.

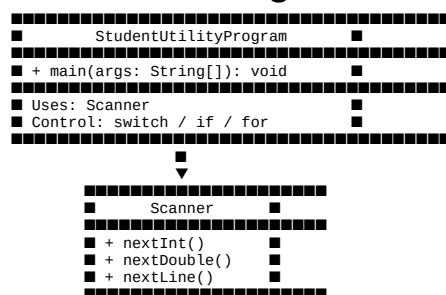
System Requirements

HARDWARE • Standard PC/Laptop • Intel Core i3/i5 processor or equivalent • Minimum 4 GB RAM • Minimum 500 MB free storage	SOFTWARE • Operating System: Windows / Linux / macOS • Java Development Kit (JDK 17 or JDK 21) • Text Editor / IDE: Eclipse, IntelliJ IDEA, NetBeans, or VS Code • Command Prompt / Terminal
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Algorithm / Logical Description

1. Import Scanner for keyboard input.
2. Display four main parts and read the user's choice.
3. Use an outer switch to select Part 1–Part 4.
4. Part 1 performs student information, marks entry, total and percentage calculation.
5. Part 2 performs even/odd checking, largest of three numbers, grading, and day selection.
6. Part 3 performs multiplication table, 1-to-N display, sum of N numbers, and Fibonacci series.
7. Part 4 performs factorial, prime checking, maximum of two numbers, and circle-area calculation.
8. Display the result of the selected operation.

UML Class Diagram



Relationship: StudentUtilityProgram uses the Scanner class for user input. The main method controls the program using switch statements, conditions, and loops.

Source Code

```
import java.util.Scanner;

// Main class of the program
public class StudentUtilityProgram {

    // Main method - program execution starts here
    public static void main(String[] args) {

        // Creating Scanner object for user input
        Scanner sc = new Scanner(System.in);

        // Displaying main menu
        System.out.println("1. Part 1");
        System.out.println("2. Part 2");
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System.out.println("3. Part 3");
System.out.println("4. Part 4");

// Taking main menu choice from the user
System.out.print("Enter your choice: ");
int choice = sc.nextInt();

// Main switch to select different parts
switch (choice) {

    // ===== PART 1 =====
    case 1:

        // Displaying Part 1 sub-menu
        System.out.print("Enter case 1-4 ");
        int achoice = sc.nextInt();

        // Variables for student details and marks
        String name = "";
        int rollno;
        double marks1, marks2, marks3, total, percentage;

        // Part 1 sub-menu
        switch (achoice) {

            // Case 1: Display fixed student information
            case 1:
                System.out.println("Student Information");
                System.out.println("Name : Sameera");
                System.out.println("Roll No : 102");
                break;

            // Case 2: Accept student details and marks
            case 2:
                System.out.println("Enter Student Name: ");

                // Reading student name
                name = sc.nextLine();

                System.out.println("Enter Roll Number: ");
                rollno = sc.nextInt();

                // Reading English marks
                System.out.println("Enter English Marks: ");
                marks1 = sc.nextDouble();

                // Reading Maths marks
                System.out.print("Enter Maths Marks: ");
                marks2 = sc.nextDouble();

                // Reading Social marks
                System.out.print("Enter Social Marks: ");
                marks3 = sc.nextDouble();
                break;

            // Case 3: Calculate total and percentage
            case 3:
                System.out.print("Enter English Marks: ");
                marks1 = sc.nextDouble();

                System.out.print("Enter Maths Marks: ");
                marks2 = sc.nextDouble();

                System.out.print("Enter Social Marks: ");
                marks3 = sc.nextDouble();

                // Calculating total marks
                total = marks1 + marks2 + marks3;

                // Calculating percentage
                percentage = (total / 300) * 100;

                // Displaying result
                System.out.println("Total = " + total);
                System.out.println("Percentage = " + percentage);
                break;

            // Case 4: Accept details and display complete result
            case 4:
                System.out.print("Enter Student Name: ");

                // Reading student name
                name = sc.nextLine();

                System.out.print("Enter Roll Number: ");
                rollno = sc.nextInt();

```

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        // Reading English marks
        System.out.print("Enter English Marks: ");
        marks1 = sc.nextDouble();

        // Reading Maths marks
        System.out.print("Enter Maths Marks: ");
        marks2 = sc.nextDouble();

        // Reading Social marks
        System.out.print("Enter Social Marks: ");
        marks3 = sc.nextDouble();

        // Calculating total marks
        total = marks1 + marks2 + marks3;

        // Calculating percentage
        percentage = (total / 300) * 100;

        // Displaying complete result
        System.out.println("\n----- RESULT -----");
        System.out.println("Name : " + name);
        System.out.println("Roll No : " + rollno);
        System.out.println("Total : " + total);
        System.out.println("Percentage : " + percentage);
        break;

        // Invalid Part 1 choice
        default:
            System.out.println("Invalid Choice");
    }

    break;

// ===== PART 2 =====
case 2:

    // Displaying Part 2 sub-menu
    System.out.println("enter case 1-4");
    int bchoice = sc.nextInt();

    // Part 2 sub-menu
    switch (bchoice) {

        // Case 1: Check whether a number is even or odd
        case 1:
            System.out.print("Enter a number: ");
            int n = sc.nextInt();

            // Checking divisibility by 2
            if (n % 2 == 0)
                System.out.println("Even Number");
            else
                System.out.println("Odd Number");

            break;

        // Case 2: Find the largest of three numbers
        case 2:
            System.out.print("Enter first number: ");
            int a = sc.nextInt();

            System.out.print("Enter second number: ");
            int b = sc.nextInt();

            System.out.print("Enter third number: ");
            int c = sc.nextInt();

            // Comparing three numbers
            if (a >= b && a >= c)
                System.out.println("Largest = " + a);
            else if (b >= a && b >= c)
                System.out.println("Largest = " + b);
            else
                System.out.println("Largest = " + c);

            break;

        // Case 3: Find grade based on percentage
        case 3:
            System.out.print("Enter percentage: ");
            double per = sc.nextDouble();

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        // Checking percentage and assigning grade
        if (per >= 90)
            System.out.println("Grade A");
        else if (per >= 75)
            System.out.println("Grade B");
        else if (per >= 60)
            System.out.println("Grade C");
        else if (per >= 40)
            System.out.println("Grade D");
        else
            System.out.println("Fail");

        break;

// Case 4: Display day based on day number
case 4:
    System.out.print("Enter day number (1-7): ");
    int day = sc.nextInt();

    // Nested switch for day selection
    switch (day) {

        case 1:
            System.out.println("Monday");
            break;

        case 2:
            System.out.println("Tuesday");
            break;

        case 3:
            System.out.println("Wednesday");
            break;

        case 4:
            System.out.println("Thursday");
            break;

        case 5:
            System.out.println("Friday");
            break;

        case 6:
            System.out.println("Saturday");
            break;

        case 7:
            System.out.println("Sunday");
            break;

        // Invalid day number
        default:
            System.out.println("Invalid Day");
    }

    break;
}

// ===== PART 3 =====
case 3:

    // Displaying Part 3 sub-menu
    System.out.println("enter case 1-4");
    int cchoice = sc.nextInt();

    // Part 3 sub-menu
    switch (cchoice) {

        // Case 1: Display multiplication table
        case 1:
            System.out.print("Enter a number: ");
            int num = sc.nextInt();

            System.out.println("Multiplication Table:");

            // Loop from 1 to 10
            for (int i = 1; i <= 10; i++) {
                System.out.println(
                    num + " x " + i + " = " + (num * i)
                );
            }

            break;
    }
}

```

```

// Case 2: Display numbers from 1 to N
case 2:
    System.out.print("Enter N: ");
    int n = sc.nextInt();

    System.out.println(
        "Numbers from 1 to " + n + ":"
    );

    // Printing numbers from 1 to N
    for (int i = 1; i <= n; i++) {
        System.out.print(i + " ");
    }

    break;

// Case 3: Calculate sum from 1 to N
case 3:
    System.out.print("Enter N: ");
    int m = sc.nextInt();

    // Variable to store sum
    int sum = 0;

    // Adding numbers from 1 to N
    for (int i = 1; i <= m; i++) {
        sum = sum + i;
    }

    // Displaying sum
    System.out.println("Sum = " + sum);

    break;

// Case 4: Generate Fibonacci series
case 4:
    System.out.print("Enter number of terms: ");
    int terms = sc.nextInt();

    // Initial Fibonacci values
    int a = 0, b = 1, c;

    System.out.println("Fibonacci Series:");

    // Generating Fibonacci series
    for (int i = 1; i <= terms; i++) {

        // Display current term
        System.out.print(a + " ");

        // Calculate next term
        c = a + b;

        // Update Fibonacci values
        a = b;
        b = c;
    }

    break;
}

// ===== PART 4 =====
case 4:

    // Displaying Part 4 sub-menu
    System.out.println("enter case 1-4");
    int dchoice = sc.nextInt();

    // Part 4 sub-menu
    switch (dchoice) {

        // Case 1: Calculate factorial
        case 1:
            System.out.print("Enter a number: ");
            int n = sc.nextInt();

            // Initial factorial value
            int fact = 1;

            // Calculating factorial using loop
            for (int i = 1; i <= n; i++) {
                fact = fact * i;
            }

```

```

    }

    // Displaying factorial
    System.out.println("Factorial = " + fact);

    break;

// Case 2: Check whether a number is prime
case 2:
    System.out.print("Enter a number: ");
    int num = sc.nextInt();

    // Variable to count factors
    int count = 0;

    // Checking all possible divisors
    for (int i = 1; i <= num; i++) {

        if (num % i == 0) {
            count++;
        }
    }

    // A prime number has exactly two factors
    if (count == 2)
        System.out.println("Prime Number");
    else
        System.out.println("Not a Prime Number");

    break;

// Case 3: Find maximum of two numbers
case 3:
    System.out.print("Enter first number: ");
    int a = sc.nextInt();

    System.out.print("Enter second number: ");
    int b = sc.nextInt();

    // Comparing two numbers
    if (a > b)
        System.out.println("Maximum = " + a);
    else
        System.out.println("Maximum = " + b);

    break;

// Case 4: Calculate area of a circle
case 4:
    System.out.print("Enter radius: ");
    double r = sc.nextDouble();

    // Calculating area of circle
    double area = 3.14 * r * r;

    // Displaying area
    System.out.println(
        "Area of Circle = " + area
    );

    break;

// Invalid Part 4 choice
default:
    System.out.println("Invalid Choice");
}

break;
}

// Closing Scanner object
sc.close();
}
}

```

Completion and Execution Commands

Save the source code exactly as:

StudentUtilityProgram.java

Compile:

javac StudentUtilityProgram.java

Run:

java StudentUtilityProgram

Output Screen

```
1. Part 1
2. Part 2
3. Part 3
4. Part 4
Enter your choice: 1
Enter case 1-4 1
Student Information
Name : Sameera
Roll No : 102
```

Results

The Student Utility Program was successfully implemented in Java. The program demonstrates Scanner input, switch-case selection, nested switch statements, conditional logic, loops, arithmetic calculations, and output display.